Lab 1: Introduction to Linux – Detailed Commands, Descriptions & Examples

Lab Objectives
After completing this lab, students will be able to:
- Navigate the Linux file system
- Create, delete, copy, and move files/directories
- View file contents
- Understand file permissions
- Use help and manual systems

This lab supports foundational OS concepts discussed in Modern Operating Systems.

Basic Navigation Commands

1) pwd — Print Working Directory
Description:
Displays the full path of the current directory.

Syntax:
pwd
Example:
pwd
Output:
/home/student
Purpose:
Helps identify your current location in the Linux file system.

2) ls — List Directory Contents
Description:
Displays files and directories inside the current directory.

Syntax:
ls
Example:
ls

Output:
file1.txt file2.txt Documents

Important Options:
ls -l   : Long listing format
ls -a   : Show hidden files
ls -lh : Human-readable file size
ls -R   : Recursive listing

Example:
ls -l
Output shows:

- Permissions
- Owner
- File size
- Date modified

3) cd — Change Directory
Description:
Used to move between directories.

Syntax:
cd directory_name
Examples:
cd Documents
cd ..
cd ~
cd /

cd .. : Move one level up
cd ~ : Go to home directory
cd / : Go to root directory

File & Directory Management Commands

4) mkdir — Make Directory
Description:
Creates a new directory.

Syntax:
mkdir directory_name
Example:
mkdir os_lab

5) rmdir — Remove Empty Directory
Description:
Deletes an empty directory.

Syntax:
rmdir directory_name
Example:
rmdir os_lab
Note: Works only if the directory is empty.

6) rm — Remove Files or Directories
Description:
Deletes files or directories.

Syntax:
rm filename
rm -r directory

Examples:
rm file1.txt
rm -r os_lab

Warning: Dangerous command (no recycle bin).

7) touch — Create Empty File

Description:
Creates an empty file or updates the timestamp.

Syntax:
touch filename
Example:
touch test.txt

8) cp — Copy Files

Description:
Copies a file or directory.

Syntax:
cp source destination
Examples:
cp file1.txt file2.txt
cp -r dir1 dir2

9) mv — Move or Rename Files

Description:
Moves the file or renames it.

Syntax:
mv oldname newname
Examples:
mv file1.txt file2.txt
mv file1.txt Documents/

Viewing File Contents

10) cat — Display File Content

Description:
Displays entire file content.

Syntax:
cat filename
Example:
cat test.txt

11) less — View File Page by Page

Description:
Allows scrolling inside a file.

Syntax:
less filename
Controls:
q : quit
space : next page
/word : search

12) head — View First Lines

Syntax:
head filename
Example:
head -5 test.txt

Shows first 5 lines.

13) tail — View Last Lines

Syntax:
tail filename
Example:
tail -3 test.txt

Shows the last 3 lines.

File Permissions

14) chmod — Change File Permission

Description:
Modifies file access permissions.

Syntax:
chmod permissions filename
Permission Structure:
```
rwx    rwx    rwx
 |      |      |
user group others
```

Numeric Values:
r = 4
w = 2
x = 1

Examples:
chmod 644 file.txt
chmod 755 script.sh

15) chown — Change Owner

Syntax:
sudo chown user filename

Example:
sudo chown student file.txt

Help & Manual Commands

16) man — Manual Pages

Description:
Displays full documentation.

Syntax:
man command
Example:
man ls

Press q to exit.

17) clear — Clear Terminal Screen

Syntax:
clear

Practice Tasks for Students

Task 1:
Create a directory os_lab and inside it:
- Create 3 files
- Copy one file
- Rename one file

Task 2:
- Change the permission of one file to read-only
- Display the last 2 lines of a file

Task 3:
Create nested directories using:
mkdir -p dir1/dir2/dir3